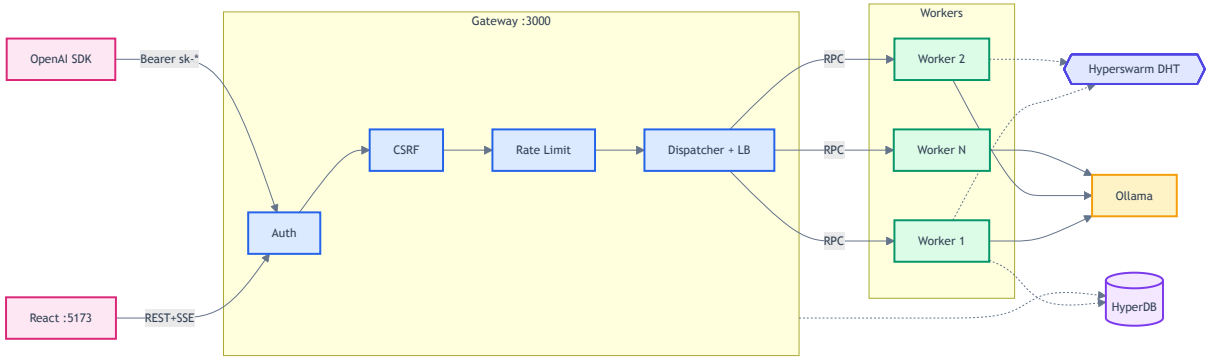
AI PaaS — Architecture Overview

GPU Cloud Service Platform • Distributed P2P • OpenAI Compatible • March 2026

AI PaaS is a distributed GPU cloud platform providing API key management, model deployment, and OpenAI-compatible inference. Workers are discovered via Hyperswarm DHT — no central coordinator. State is replicated across all nodes via HyperDB.

System Architecture



Client → Gateway (auth, rate limit) → Workers (RPC) → Ollama → stream back

Request Flows

- **Inference:** Client → Gateway → RPC → Worker → Ollama → SSE stream back
- **Deploy:** Client → Gateway → Worker `/stream/pull` → Ollama pull → SSE progress
- **Streaming:** Gateway proxies SSE from worker HTTP, manages job slots

Design Principles

- No shared singletons — DI everywhere
- Swappable strategies — LB, rate-limit, runtime
- P2P-first — Hyperswarm auto-discovery
- OpenAI compatible — any SDK works

Technology Stack

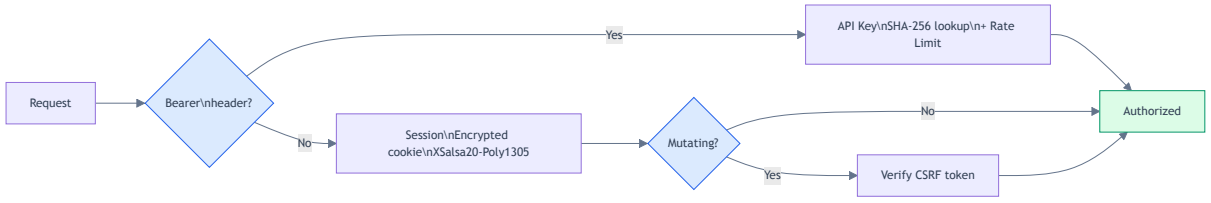
LAYER	TECHNOLOGY	LAYER	TECHNOLOGY
Runtime	Node.js 22	Frontend	React + Vite + Zustand
Framework	Fastify v5	Server State	TanStack React Query
Worker Comms	Hyperswarm RPC	Charts	Recharts
Data Store	HyperDB + Hypercore	Auth	@fastify/secure-session
Inference	Ollama (swappable)	CSRF	@fastify/csrf-protection
Streaming	@fastify/sse	Tooling	Biome + pnpm

Data Layer — HyperDB

Distributed state replicated across all nodes. No central database.

COLLECTION	KEY	INDEXES	PURPOSE
`apiKeys`	`id`	`by-hash` (unique)	O(1) key validation via SHA-256
`deployments`	`id`	`by-model`	Find deployments per model
`usageRecords`	`id`	`by-keyId`, `by-model`, `by-timestamp`	Usage lookup + range queries (last 24h aggregation)

Authentication & CSRF



Dual auth: session cookies (dashboard) + API keys (programmatic). CSRF skipped for Bearer.

API Key Security

- `sk-` + base62(randomBytes(32)) — 256-bit
- Stored as SHA-256 hash only
- View-once: full key shown once at creation
- Display: `sk-a8Kx••••` after creation

Session Security

- XSalsa20-Poly1305 encrypted cookies
- `session.regenerate()` on login
- CSRF token seeded via `generateCsrf()`
- httpOnly, secure, sameSite: strict

Rate Limiting

Composite pattern — **all strategies must pass**. Per-key thresholds stored in HyperDB.

STRATEGY	ALGORITHM	DEFAULT LIMIT
TokenBucket	Refillable bucket, burst-friendly	60 requests/min
SlidingWindow	Timestamp log, filters outside window	100K tokens/hour
FixedWindow	Time-bucketed counters	Configurable

Load Balancing

STRATEGY	LOGIC	BEST FOR
ModelAffinity	Prefer worker with model loaded	Avoid cold starts
LeastConnections	Fewest active jobs	Heterogeneous load
RoundRobin	Circular rotation	Even distribution

Worker Architecture

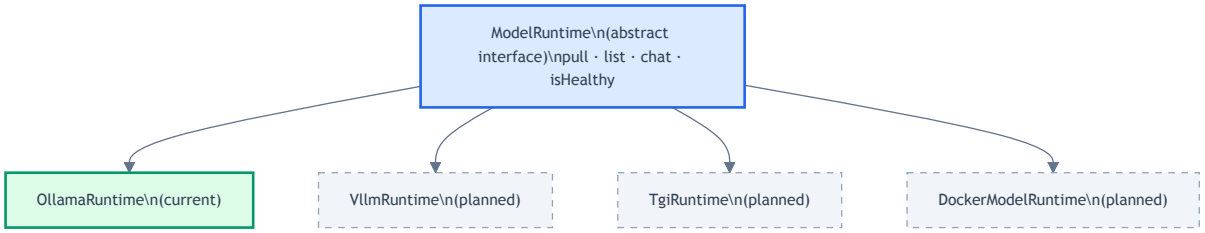
RPC Methods (Hyperswarm)

- `health.check` — heartbeat (30s, 3-strike)
- `inference.chat` — non-streaming
- `model.pull/list/delete/show`

HTTP Streaming (SSE)

- `/stream/chat` — token-by-token
- `/stream/pull` — pull progress %
- Job slots: acquire before, release in finally

Model Runtime Adapter (Swappable)



MODEL_RUNTIME=ollama env var selects adapter. No worker code changes when swapping.

Docker Topology

SERVICE	PORT	ROLE
dht-bootstrap	49737	DHT bootstrap node for peer discovery
gateway	3000	Fastify API — auth, routing, SSE, OpenAPI docs
worker-1, worker-2	8080 (internal)	RPC server + HTTP stream server
frontend	5173	Vite dev server — React dashboard
Ollama	11434 (host)	Model inference via `host.docker.internal`

> **Note:** Base image is `node:22-slim` (not Alpine — `udx-native` lacks musl prebuild). Source bind mounts for hot reload. Anonymous `node_modules` volumes require `--renew-anon-volumes` on dep changes.

Scaling & Production Path

CURRENT	PRODUCTION	WHY
Ollama	vLLM / TGI	Continuous batching, PagedAttention
HyperDB	PostgreSQL + Redis	Strong consistency, Redis rate-limit hot path
Single gateway	Gateway cluster + envoy	HA, TLS termination, connection pooling
EnvUserService	DB-backed users	Multi-user, roles, teams

FAILURE	DETECTION	MITIGATION
Worker crash	Health check (3 failures)	Exclude from LB, 503 + Retry-After
Gateway crash	Docker restart	`recoverStuckDeployments()` on boot
Stream disconnect	`sse.isConnected`	`releaseOnce()` guard frees job slot